Zoho Schools for Graduate Studies

Notes

Interface

- An Interface is defined using the keyword interface.
- Interface Name begins with uppercase.

Implements

- It will inherit all the members in the interface.
- It has to provide implementation for abstract method defined in the interface.

Prior to Java 8 :

- An interface is an 100% abstract class, .i.e. it contains only abstract Methods.
- It can have constants.

From java 8:

- It can contain abstract methods.
- It can contain only those concrete method that are marked as default, static, private.
- It can have constants.

In Java, interfaces can have data members, but they are implicitly public, static, and final. This means they are effectively constants.

Here's what that implies:

Public: These data members are accessible from anywhere.

Static: They belong to the interface itself, not to any specific instance of a class implementing the interface. You access them using the interface name (e.g. MyInterface.MY_CONSTANT).

Final: Their values cannot be changed after initialization. They must be assigned a value when declared.

Note:

- By default, method declarations inside an interface are abstract & public.
- By default variables declaredinside an interface are public, static & final.
- An interface can inherit (extend) multiple interfaces.
- An interface cannot implement another interface
- A class can extends only one another class except object class.
- A class can implements multiple interfaces.
- A class cannot extends another interfaces.
- A class cannot implement another class.

1) Who's responsibility to provide implementation for abstract method?

Ans: Subclass

2) Why error occurred ?

```java
package part2;

public interface Display {
    final int x=5;
    void studentDetails(); // abstract method
}
```

```java
package part2;

public interface Display {
    final int x;
    void studentDetails(); // abstract metho
}
```

Ans: Being a constant we should initialize it at the place of declaration.

3) why error occurred ?

Ans:

```java
package part2;

public class DisplayDemo implements Display{
    void studentDetails() {
        
    }
    public static void main(String[] args) {
        // TODO Auto-generated method stub

    }

}
```

1. By default, method declarations inside an interface are abstract & public.
2. Rule of inheritance , overloaded methods can share same access level or can have a broader access level.(In interface the jvm will provide public as access level for abstract method).

4) A class implements interface but unwilling to provide implementation for those abstract methods then what can do?

Ans: To declare the class under abstract .

InterfaceDemo.java Display.java *DisplayDemo.java

```java
package part2;

abstract public class DisplayDemo implements Display{
    /*public void studentDetails() {
        System.out.println("Student details...");
    }*/
    public static void main(String[] args) {
        DisplayDemo d=new DisplayDemo();
        d.studentDetails();

    }
}
```

5) Attributes that declare inside interface when they get memory allocation and Why those members are static?

Ans: At the time of class Loading the memory will allocate to the attributes.